

# Møller Polarimetry on Atomic Hydrogen

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PAVI11 Rome

- 1 Motivation for precise polarimetry
- 2 Møller Polarimetry
- 3 Møller with Atomic Hydrogen Target

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# PV opportunities at 11-GeV

CEBAF is an excellent facility for PV experiments

## PV at 6 GeV

- High polarization  $\sim 85\%$
- High beam current  $< 100\mu\text{A}$
- Low noise beam

Measured:  $G_s$

Elastic  $e p$ ,  $e {}^4\text{He}$  (HAPPEX, G0);

EW  $e d$  DIS;

Running:

- Neutron skin  ${}^{208}\text{Pb}$   
 $e \text{Pb} \rightarrow e \text{Pb}$  (PREX)
- EW  $e p \rightarrow e p$  (QWEAK)

## PV at 11 GeV

- Same polarization
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Higher energies:

$A \propto Q^2$  larger, but

$\sigma_{\text{elastic}}$  suppressed by FF

Approved by PAC:

- PV Møller
- PV DIS (SoLID)

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# 11 GeV JLab: Motivation for precise polarimetry

## PV Møller

| Source of error      | % error |
|----------------------|---------|
| $Q^2$ absolute value | 0.5     |
| beam polarization    | 0.4     |
| beam second order    | 0.4     |
| inelastic $ep$       | 0.4     |
| elastic $ep$         | 0.3     |
| other                | 0.5     |
| total                | 1.0     |

## PV DIS (with SoLID)

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|-----------------------|---------|
| beam polarization     | 0.4     |
| radiative corrections | 0.3     |
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| statistics            | 0.3     |
| total                 | 0.6     |

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# Electron Polarimetry for JLab at 11 GeV: Features

- Energy range  $E_{beam} = 2.0 - 11 \text{ GeV}$
  - Current range  $E_{beam} = 50 - 80 \mu\text{A}$
  - Beam pulse  $\sigma_t = 0.5 \text{ ps}$ ,  $\sigma_R \sim 100 \mu\text{m}$  repetition rate  $500 \text{ MHz}$
- 
- Statistical error for a period of a possible polarization change ( $\sim 1 \text{ h}$ )
  - Systematic error
    - Does polarimetry use the same beam (energy, current, location) as the experiment?
    - Continuous or intermittent (invasive?)

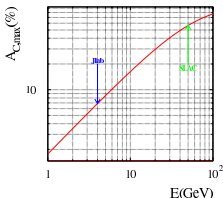
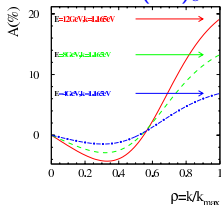
# Compton Polarimetry

$$\frac{\sigma_{\uparrow\uparrow} - \sigma_{\uparrow\downarrow}}{\sigma_{\uparrow\uparrow} + \sigma_{\uparrow\downarrow}} = A \cdot \mathcal{P}_b \mathcal{P}_t$$

# Møller Polarimetry

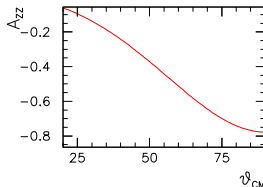
$$\tilde{e}^- + (h\nu)_\sigma \rightarrow e^- + \gamma \text{ QED.}$$

$$\tilde{e}^- + \tilde{e}^- \rightarrow e^- + e^- \text{ QED.}$$



- Rad. corrections to Born  $< 0.1\%$
- Detecting:  $\gamma$  ( $0^\circ$ ),  $e^-$   $E < E_0$
- Strong  $\frac{dA}{dk'}$  - good  $\sigma E_\gamma / E_\gamma$  needed
- $A \propto kE$  at  $E < 20$  GeV
- $T \propto 1/(\sigma \cdot A^2) \propto 1/k^2 \times 1/E^2$
- $\mathcal{P}_{laser} \sim 100\%$
- Non-invasive measurement

Syst. error 3→50 GeV:  $\sim 1. \rightarrow 0.5\%$



$$A(E) = -\frac{7}{9} \sigma_{lab} \sim 180 \frac{mb}{ster}$$

- Rad. corrections to Born  $< 0.3\%$
- Detecting the  $e^-$  at  $\theta_{CM} \sim 90^\circ$
- $\frac{dA}{d\theta_{CM}}|_{90^\circ} \sim 0$  - good systematics
- Beam energy independent
- Coincidence - no background
- Ferromagnetic target  $\mathcal{P}_T \sim 8\%$ 
  - $\langle I_B \rangle < 5 \mu A$  (heating)
  - Levchuk effect
  - Low  $\mathcal{P}_T \Rightarrow$  dead time
  - Syst. error  $\sigma(\mathcal{P}_T) > 0.4\%$

# Possible Breakthrough in Accuracy

Møller polarimetry with 100% polarized atomic hydrogen gas, stored in a ultra-cold magnetic trap.

*E.Chudakov and V.Luppov IEEE Trans. on Nucl. Sc., 51, 1533 (2004)*

[http://www.jlab.org/~gen/hyd/loi\\_3.pdf](http://www.jlab.org/~gen/hyd/loi_3.pdf)

## Advantages:

- 100% electron polarization
  - very small error on polarization
  - sufficient rates  $\sim \times 0.005$  - no dead time
  - false asymmetries reduced  $\sim \times 0.1$
- Hydrogen gas target
  - no Levchuk effect
  - low single arm BG from rad. Mott ( $\times 0.1$  of the BG from Fe)
  - high beam currents allowed: continuous measurement

## Operation:

- density:  $\sim 6 \cdot 10^{16}$  atoms/cm<sup>2</sup>
- Stat. error at 50  $\mu$ A: 1% in  $\sim 10$  min

# Møller Systematic Errors

Proposed: 100%-polarized atomic hydrogen target ( $\sim 3 \cdot 10^{16}$  atoms/cm<sup>2</sup>).

| Variable            | Hall C | Hall A  |         |          |
|---------------------|--------|---------|---------|----------|
|                     |        | Present | Upgrade | Proposed |
| Target polarization | 0.25%  | 2.00%   | 0.63%*  | 0.01%    |
| Target angle        | 0.00%  | 0.50%   | 0.00%   | 0.00%    |
| Analyzing power     | 0.24%  | 0.30%   | 0.30%   | 0.10%    |
| Levchuk effect      | 0.30%  | 0.20%   | 0.50%   | 0.00%    |
| Target temperature  | 0.05%  | 0.00%   | 0.02%   | 0.00%    |
| Dead time           | -      | 0.30%   | 0.30%   | 0.10%    |
| Background          | -      | 0.30%   | 0.30%   | 0.10%    |
| Others              | 0.10%  | 0.30%   | 0.30%   | 0.30%?   |
| Total               | 0.47%  | 2.10%   | ~1.00%  | ~0.35%   |

\*  $0.25 \oplus 0.5$  (Hall A different targets)  $\oplus 0.3$  (Hall A saturation effects)

# Hydrogen Atom in Magnetic Field

$$H_1: \vec{\mu} \approx \vec{\mu}_e;$$

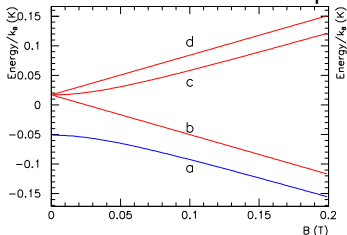
$H_2$ : opposite electron spins

Consider  $H_1$  in  $B = 7 \text{ T}$  at  $T = 300 \text{ mK}$

At thermodynamical equilibrium:

$$n_+/n_- = \exp(-2\mu B/kT) \approx 10^{-14}$$

Complication from hyperfine splitting:



Low energy

$$|b\rangle = |\downarrow\downarrow\rangle$$

$$|a\rangle = |\downarrow\uparrow\rangle \cdot \cos\theta - |\uparrow\uparrow\rangle \cdot \sin\theta$$

High energy

$$|d\rangle = |\uparrow\uparrow\rangle$$

$$|c\rangle = |\uparrow\downarrow\rangle \cdot \cos\theta + |\downarrow\downarrow\rangle \cdot \sin\theta$$

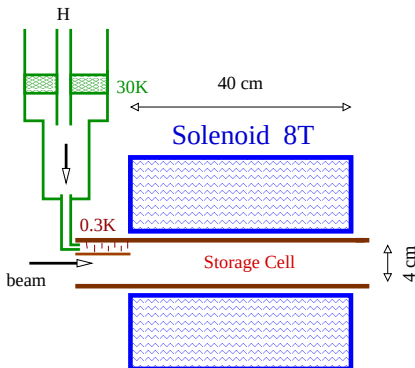
where  $\tan 2\theta \approx 0.05/B(T)$ , at 7 T  $\sin\theta \approx 0.0035$

Mixture  $\sim 53\%$  of  $|a\rangle$  and  $\sim 47\%$  of  $|b\rangle$ :

$$\mathcal{P}_e \sim 1 - \delta, \quad \delta \sim 10^{-5},$$

$$\mathcal{P}_p \sim -0.06 \text{ (recombination)} \Rightarrow \sim 80\%$$

# Storage Cell



First: 1980 (I.Silvera,J.Walraven)  
 $\vec{p}$  jet (Michigan)  
 Never put in high power beam

- $-\vec{\nabla}(\vec{\mu}_H \vec{B})$  force in the field gradient
  - pulls  $|a\rangle, |b\rangle$  into the strong field
  - repels  $|c\rangle, |d\rangle$  out of the field
- $H+H \rightarrow H_2$  recombination (+4.5 eV)  
 high rate at low T
  - parallel electron spins: suppressed
  - gas: 2-body kinematic suppression
  - gas: 3-body density suppression
  - surface: strong unless coated  
 $\sim 50$  nm of superfluid  $^4\text{He}$
- Density  $3 \cdot 10^{15} - 3 \cdot 10^{17} \text{ cm}^{-3}$ .
- Gas lifetime  $> 1 \text{ h}$ .

# Dynamic Equilibrium and Proton Polarization

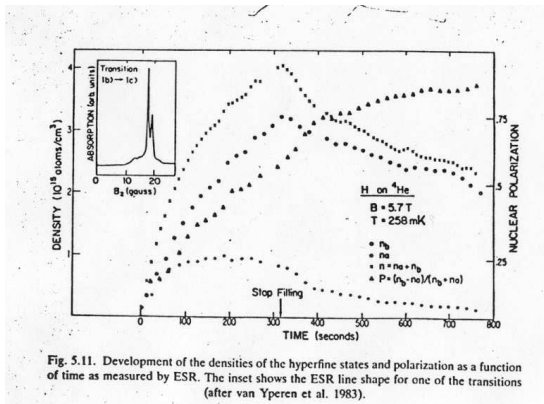
Proton polarization builds up, because of recombination of states with opposite electron spins:

$$|a\rangle = |\downarrow\uparrow\rangle\alpha + |\uparrow\downarrow\rangle\beta \text{ and}$$

$$|b\rangle = |\downarrow\downarrow\rangle$$

As a result,  $|a\rangle$  dies out and only  $|b\rangle = \downarrow\downarrow$  is left!

$$P \rightarrow 0.8$$



# Contaminations and Depolarization of the Target Gas

Ideally, the trapped gas polarization is nearly 100% ( $\sim 10^{-5}$  contamination).  
Good understanding of the gas properties (without beam).

## Gas Properties

- Atom velocity  $\approx 80$  m/s
- Atomic collisions\*  $\approx 1.4 \cdot 10^5$  s $^{-1}$
- Mean free path  $\lambda \approx 0.6$  mm
- Wall collision time  $t_R \approx 2$  ms
- Escape (10cm drift)  $t_{es} \approx 1.4$  s

## CEBAF Beam

- Bunch length  $\sigma = 0.5$  ps
- Repetition rate 497 MHz
- Beam spot diameter  $\sim 0.2$  mm

\* Based on  $\sigma_{HH} = 42 \cdot 10^{-16}$  cm $^2$   
needs verification

## Contamination and Depolarization

### No Beam

- Hydrogen molecules  $\sim 10^{-5}$
- Upper states  $|c\rangle$  and  $|d\rangle < 10^{-5}$
- Excited states  $< 10^{-5}$
- Helium and residual gas  $< 0.1\%$   
- measurable with the beam

### 100 $\mu$ A Beam

- Depolarization by beam RF  $< 2 \cdot 10^{-4}$
- Ion, electron contamination  $< 10^{-5}$
- Excited states  $< 10^{-5}$
- Ionization heating  $< 10^{-10}$

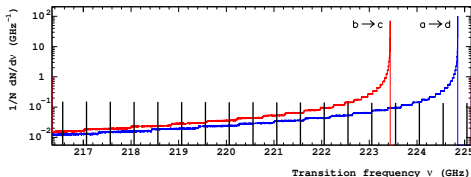
Expected depolarization  $< 2 \cdot 10^{-4}$

# Contaminations and Depolarization of the Target Gas

100  $\mu\text{A}$  CEBAF beam:

## Beam RF influence

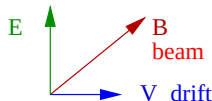
- $|a\rangle \rightarrow |d\rangle$  and  $|b\rangle \rightarrow |c\rangle \sim 200 \text{ GHz}$
- RF spectrum: flat at  $< 300 \text{ GHz}$



- $\sim 10^{-4} \text{ s}^{-1}$  conversions (all atoms)
- $\sim 6\% \text{ s}^{-1}$  conversions (beam area)
- Diffusion: contamination  
 $\sim 1.5 \cdot 10^{-4}$  in the beam area
- Solenoid tune to avoid resonances

## Gas Ionization

- $10^{-5} \text{ s}^{-1}$  of all atoms
- $20\% \text{ s}^{-1}$  in the beam area
- Problems:
  - No transverse diffusion
  - Recombination suppressed
  - Contamination  $\sim 40\%$  in beam
- Solution: electric field  $\sim 1 \text{ V/cm}$ 
  - Drift  $\vec{v} = \vec{E} \times \vec{B} / B^2 \sim 12 \text{ m/s}$
  - Cleaning time  $\sim 20 \mu\text{s}$
  - Contamination  $< 10^{-5}$
  - Ions, electrons: same direction
  - Beam  $\vec{E}_r(160\mu\text{m}) \approx 0.2 \text{ V/cm}$



# Summary on Atomic Hydrogen for Møller Polarimetry

## Potential for Polarimetry

- Systematic accuracy of  $< 0.3\%$
- Continuous measurements
- Tools for systematic studies:
  - changing the electrical field (ionization)
  - changing the magnetic field (RF depolarization)

## Problems and Questions

- Electrodes in the cell: R&D is needed
- Residual gas  $0.1\%$  accurate subtraction
  - Coordinate detectors: the interaction point?
- Atomic cross section (mean free path...) needs verification
- Cost and complexity

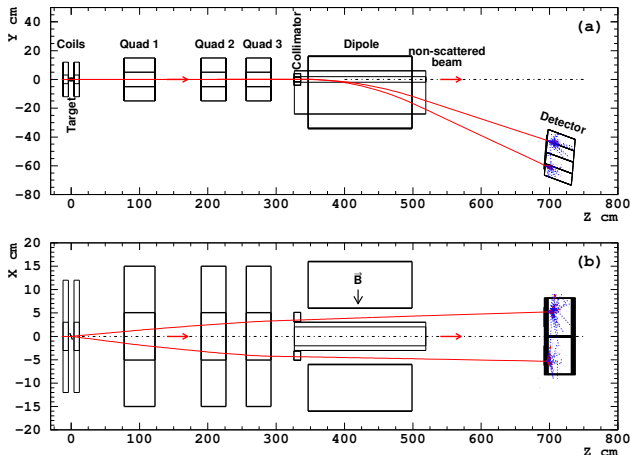
# Conclusion

New PV experiments require a  $\sim 0.4\%$  polarimetry.

## Possible strategy

- Continuous polarimetry
- $\sim 0.4\%$  Compton - seems feasible
- Second polarimetry method (Møller) of similar accuracy would help dramatically.
- Møller with iron targets - rather 1% than 0.4%
- Møller with atomic hydrogen target potentially can provide  $< 0.30\%$  accuracy, very complex, needs R&D

# Appendix: Hall A Møller Polarimeter with Saturated Iron Target



Upgraded to saturated foils (as in Hall C).

- Minimal Levchuk
- $\sigma_{stat} = 1\%$  in  $\sim 2-3$  min
- $B_z \sim 4$  T field
- Foil at  $0^\circ$  to field
- Foils  $1-10\mu\text{m}$
- Beam  $< 5\mu\text{A}$
- Syst. error  $\sim 1\%$

# Appendix: Gas Properties

- $n = 2 \cdot 10^{15} \text{ cm}^{-3}$  - density
- $T = 0.3 \text{ K}$  - temperature
- Diffusion speed  $\Rightarrow$  cleaning time
- Heat conductance
- Depend on the atomic cross-section  $\sigma$

| Ref.,<br>date | condi-<br>tions      | H polarized                          |                              | H unpolarized                        |                              |
|---------------|----------------------|--------------------------------------|------------------------------|--------------------------------------|------------------------------|
|               |                      | $\sigma, \text{ cm}^2$<br>$10^{-16}$ | $d, \text{ cm}$<br>$10^{-8}$ | $\sigma, \text{ cm}^2$<br>$10^{-16}$ | $d, \text{ cm}$<br>$10^{-8}$ |
| Allison,71    | $T > 1 \text{ K}$    | 87.0                                 | 5.26                         | 68.0                                 | 4.65                         |
| Miller,77     | $T \sim 0 \text{ K}$ | 42.3                                 | 3.69                         | -                                    | -                            |
| Friend,80     | $T \sim 0 \text{ K}$ | 6.5                                  | 1.44                         | 4.9                                  | 1.25                         |
| Lhuillier,83  | $T = 2.5 \text{ K}$  | $\sim 30.0$                          | 3.10                         | -                                    | -                            |

Using Miller,77:

- $\bar{v} = \sqrt{8kT/\pi m} = 80 \text{ m/s}$  - atom speed
- $\frac{dn_{col}}{dt} = \sigma \cdot 4n\sqrt{\frac{kT}{\pi m}} \approx 1.4 \cdot 10^5 \text{ s}^{-1}$  - atomic collisions
- $\ell = (\sigma n \sqrt{2})^{-1} \approx 0.57 \text{ mm}$  - mean free path
- $\tau_{es} \approx 1.4 \text{ s}$  - mean drift time to  $|Z| = 10 \text{ cm}$
- $\tau_R \approx 2 \text{ ms}$  - mean drift time  $R=0 \rightarrow R=2 \text{ cm}$

# Appendix: CEBAF Beam Parameters

## General

- $\tau = 0.5 \text{ ps}$  - bunch time width (RMS) in LAB frame
- $\sigma_{Bx/y} = 100 \text{ } \mu\text{m}$  - bunch transverse width (RMS)
- $\mathcal{F} = 497 \text{ MHz}$  - bunch repetition rate
- $\gamma \geq \sim 10^4$  - beam  $\gamma$ -factor
- $\mathcal{I}_b = 100 \text{ } \mu\text{A}$  - average beam current

## Electromagnetic Field of the Bunch

In CM of the bunch:  $\sigma_z > 15 \text{ cm} \gg R_{\text{pipe}} \Rightarrow E_B \propto r^{-1}$ . Boost to Lab.

- The field is located in a thin disk around the bunch
- $\vec{E}(z, r, t)$  - radial
- $\vec{B}(z, r, t)$  - azimuthal,  $B(z, r, t) = E(z, r, t)/c$
- $B(0, r, t) = \frac{\mathcal{I}_b}{\mathcal{F} \cdot \tau} \cdot e^{-0.5(t/\tau)^2} \cdot (1 - e^{-0.5(r/\sigma_{Bx})^2}) \frac{1}{r} \cdot \frac{\mu_0}{(2\pi)^{3/2}}$   
 $B(0, r, t)_{\text{max}} = B(0, 160 \mu\text{m}, 0) = 0.144 \text{ mT}$
- $W = \mathcal{F} \cdot |B|^2 c / \mu_0 / 2 \approx 2 \text{ mW}$  - energy flux  $\ll 20 \text{ mW}$  of the refrigerator.

# Appendix: Beam Spectral Density

## One Bunch

$$B(t) = B_o \cdot \exp(-\frac{t^2}{2\tau^2}) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\omega \cdot \hat{B}(\omega) e^{i\omega t}, \text{ where}$$

$$\hat{B}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt \cdot f(t) e^{-i\omega t} = B_o \cdot \tau \cdot \exp(-\frac{\omega^2 \tau^2}{2})$$

$$\tau \cdot 200 \text{ Ghz} \cdot 2\pi = 0.62 - \text{no spectral suppression!}$$

## RF

$$B(t) = \sum_{n=-\infty}^{\infty} \hat{B}_n \cdot e^{i\omega_o n t}, \text{ where } \omega_o = 2\pi \mathcal{F}.$$

$$\hat{B}_n \approx B_o \sqrt{2\pi} \cdot \tau \mathcal{F} \cdot \exp(-\frac{\omega_o^2 n^2 \tau^2}{2}),$$

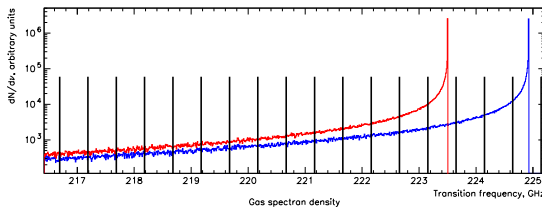
# Appendix: Depolarization by Beam RF

$|a\rangle \rightarrow |d\rangle$  and  $|b\rangle \rightarrow |c\rangle$  transitions  $\sim 200$  GHz.

$B_r$  coupling: harmonic perturbation  $\mu_e \cdot B \cdot e^{i\omega t} \Rightarrow$

$$\frac{dV_{a \rightarrow d}}{dt} = \frac{2\pi}{\hbar^2} |\mu_e \cdot B|^2 \delta(\omega - \omega_{ad})$$

One bunch:  $V_{a \rightarrow d}^B = \left(\frac{\mathcal{I}_b}{\mathcal{F}}\right)^2 \left(\frac{\mu_o \mu_e}{2\pi \hbar r_o}\right)^2 \cdot \exp(-\omega_{ad}^2 \tau^2) G^2(r)$



Diffusion:

100  $\mu A$  inflicts:

- $\sim 10^{-4} \text{ s}^{-1}$  conversions (all atoms)
- $\sim 6\% \text{ s}^{-1}$  conversions (beam area)
- Lifetime  $\tau_d \approx 1.4 \text{ s}$ , lateral diffusion  $\sim 2 \text{ ms}$
- Contamination (beam area)  $\sim 1.5 \cdot 10^{-4}$
- Solenoid uniformity  $< 10^{-3}$
- Magnetic field tune to avoid resonance

# Appendix: Beam Ionization

## Energy Loss

- 6.3 MeV/g·cm<sup>2</sup> full loss per beam particle
- 1.8 MeV/g·cm<sup>2</sup> carried out of the cell by  $\delta$ -electrons
- 4.5 MeV/g·cm<sup>2</sup> absorbed in the cell
- 100  $\mu$ A,  $3 \cdot 10^{15}$  cm<sup>-3</sup>:  
1.4 · 10<sup>13</sup> eV/s/cm loss  $\Rightarrow$  0.04 mW in the cell

## Gas Heating

- $\Delta T \propto \mathcal{I}_b \cdot n\sigma / \sqrt{T}$ ,
- $\Delta T(0) \approx 0.03 K \cdot \ln(\frac{R}{r_B} + 0.5)$ ,
  - $r_B = 0.2$  mm (no raster)  $\Delta T(0) \approx 0.140$  K
  - $r_B = 2.0$  mm (raster)  $\Delta T(0) \approx 0.075$  K
- Density  $n \cdot \sqrt{T} = \text{const}$  : 10% drop at the center
- Depolarization:  $10^{-16} \rightarrow 10^{-11}$  - OK
- Is raster needed?

# Appendix: Beam Ionization

## Ions and Free Electrons

Primary and secondary ionization in hydrogen: 40 eV/pair

100  $\mu\text{A}$ ,  $3 \cdot 10^{15} \text{ cm}^{-3}$ :  $3.5 \cdot 10^{11} \text{ pairs/s/cm}$  :  $10^{-5} \text{ s}^{-1}$  - all atoms,  $\sim 20\% \text{ s}^{-1}$  - beam area

Problems:

- No transverse diffusion of charged particles
- Recombination kinematically suppressed
- For  $\tau_{ch} = 1.4 \text{ s}$  contamination  $\sim 40\%$  in the beam

Solution: electric field  $\sim 1 \text{ V/cm}$  normal to the axis

- Larmor  $\omega_L = q_e B / m = 1.4 \cdot 10^{12} \text{ s}^{-1}$  for  $e^-$ ,  $0.8 \cdot 10^9 \text{ s}^{-1}$  for  $p \Rightarrow \omega_L \tau_{coll} \gg 1$
- Drift with  $v = \vec{E} \times \vec{B} / B^2 \sim 12 \text{ m/s}$
- Cleaning time  $\sim 20 \mu\text{s}$  : contamination  $< 10^{-5}$
- Positive and negative - the same direction
- Average field of the beam  $E_{max}(160 \mu\text{m}) \approx 0.2 \text{ V/cm}$  - not important
- Design for electrodes?

# Appendix: Excitations by the Beam Particles

## Neutral atoms $n > 1$

- Total energy release:  $1.4 \cdot 10^{13}$  eV/s/cm
- Excitation energy  $> 10$  eV
- Excited atoms  $< 1.4 \cdot 10^{12}$  eV/s/cm -  
 $< 4 \cdot 10^{-5}$  s $^{-1}$  of all, 50% - wrong polarization
- For  $\tau_{es} = 1.4$  s    contamination  $< 3 \cdot 10^{-5}$

## $|a\rangle \rightarrow |d\rangle$ transitions

- Single beam particle - similar to the beam RF calculation
- Integrating for  $r > 10^{-8}$  cm     $V_{a \rightarrow d}^B \sim 10^{-12}$  s $^{-1}$  : negligible